Quantitative non-saturation of scalar–Weyl functionals near negative Einstein metrics

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1. Main theorem

Let (M, g) be a closed connected oriented smooth four-manifold. Fix a Spin^c structure, with real first Chern class a, whose unperturbed Seiberg–Witten equations have a solution for every metric conformal to g. The structure can depend on g; it is fixed while the conformal weight varies. Write

$$V = \mathrm{Vol}_g(M), \qquad p = (a_g^+)^2 > 0, \qquad r = \fint_M |\mathrm{Ric} + g|^2, \qquad b_a = \frac{(-a^2)_+}{V}.$$

Ricci uses the tensor norm, two-forms the exterior-algebra norm, and W^+ the Hilbert–Schmidt norm as an endomorphism of Λ^+. Set $w = \sqrt{6}|W^+|$. For a constant $\beta > 0$, define the dimensionless, nonnegative functional deficit

$$\mathfrak{D}_\beta(g, a) = \frac{\beta + 1}{\beta} \int_M (s^2 + \beta w^2) - 72\pi^2 p. \tag{1}$$

Here dimensionless means invariant under constant metric rescaling; $\mathfrak{D}_\beta/V$, like r, is measured after fixing the Einstein target to $-g$.

Theorem 1 (functional non-saturation). For every finite $L > 2$ there is $c_L > 0$ such that, simultaneously for $2 \leqslant \beta \leqslant L$,

$$\boxed{(1 + b_a)\left(r + \frac{\mathfrak{D}_\beta(g, a)}{V}\right) \geqslant c_L(\beta - 2)^2.} \tag{2}$$

There is no scalar-curvature band, Sobolev constant, injectivity-radius, diameter, volume, or full-curvature hypothesis. Actual conformal monopole solvability and the displayed square cost are the inputs.

If the selected class additionally satisfies $a^2 \geqslant T := 2\chi(M) + 3\tau(M)$, there is $c_L' > 0$ with

$$\boxed{r + \frac{\mathfrak{D}_\beta(g, a)}{V} \geqslant c_L'(\beta - 2)^2.} \tag{3}$$

In particular, for an exact metric $\mathrm{Ric} = -g$ and such a class,

$$\int (s^2 + 6\beta|W^+|^2) \geqslant \frac{72\pi^2\beta}{\beta + 1} p + c_L' \frac{\beta}{\beta + 1} (\beta - 2)^2 V. \tag{4}$$

The endpoint $\beta = 2$ must remain: a negative Kähler–Einstein surface with its canonical Spin^c structure has $r = 0$ and $\mathfrak{D}_2 = 0$. Thus (2) is a compatibility theorem, not a prohibition of the Kähler–Einstein branch.

The proof supplies all auxiliary forms and their joint analytic budget from (1). It extends the former calibration at $\beta = 8$ to a parameter family of curvature functionals. It does not, by itself, identify a new smooth topology with no Einstein metric.

2. The exact deficit decomposition

Put

$$q = -s + w, \quad q_+ = \max(q, 0), \quad q_- = \max(-q, 0), \quad \Delta_+ = \int q_+^2 - 72\pi^2 p.$$

The signed monopole estimate gives $\Delta_+ \geqslant 0$. The relevant classical inequality follows from the rescaled equations; the positive part is needed when scalar curvature changes sign. Completing a square gives the exact identity

$$\boxed{\mathfrak{D}_\beta = \Delta_+ + \int q_-^2 + \frac{1}{\beta}\int (s + \beta w)^2.} \tag{5}$$

Indeed $(\beta + 1)(s^2 + \beta w^2)/\beta = (-s + w)^2 + (s + \beta w)^2/\beta$.

Write $e = s + 4$, $z = s + \beta w$, and

$$q_\beta = 4\frac{\beta + 1}{\beta}, \qquad \kappa = r + \mathfrak{D}_\beta / V.$$

Then

$$w - 4/\beta = (z - e)/\beta, \qquad q - q_\beta = z/\beta - (1 + 1/\beta)e, \qquad \int e^2 \leqslant 4Vr. \tag{6}$$

Consequently, uniformly for $2 \leqslant \beta \leqslant L$,

$$\int \left[(q - q_\beta)^2 + (w - 4/\beta)^2\right] \leqslant C_L V \kappa, \quad \int (|\operatorname{Ric}|^2 + s^2 + |W^+|^2) \leqslant C_L V(1 + \kappa). \tag{7}$$

These bounds control the Weyl norm, not its eigenspaces. The latter come from the actual monopole below.

3. Actual bounded spinors with no scalar band

Set

$$Q = \max(q, 3), \qquad f_0 = \max\{\sqrt{Q}, (-s)_+/4\}.$$

Choose smooth positive majorants $f_j \to f_0$ uniformly on the fixed compact manifold. Conformal covariance, applied to $\widehat{g} = f_j^{-2} g$, supplies solutions

$$D_A \Phi = 0, \quad -iF_A^+ = f_j \sigma(\Phi), \quad \psi = 2\sqrt{2}\sigma(\Phi), \quad \rho = |\Phi|^2 = |\psi|, \quad u = 3\rho/2. \tag{8}$$

We use $|\sigma(\Phi)|^2 = |\Phi|^4/8$. The spinor equation is $4\nabla_A^* \nabla_A \Phi + (s + f_j \rho)\Phi = 0$. At a maximum of ρ, $f_j \rho \leqslant -s$, so

$$0 \leqslant \rho \leqslant 4, \qquad 0 \leqslant u \leqslant 6. \tag{9}$$

Every approximation error below tends to zero for the fixed g, a. No uniform smoothing rate along a sequence and no bounds on derivatives of f_j are required.

The cutoff costs only scalar error:

$$0 \leqslant Q^2 - q_+^2 \leqslant 9e^2, \qquad 0 \leqslant f_0^2 - Q \leqslant \frac{9}{16}e^2. \tag{10}$$

For the second inequality, a nonzero difference forces $(-s)_+ > 4\sqrt{3} > 6$, whence $(-s)_+ \leqslant 3|e|$. For the first, $q < 3$ implies $s > -3$, hence $e > 1$.

Define nonnegative defects

$$D_0 = 2\int |d\rho|^2 + \int (4\rho|\nabla_A\Phi|^2 - |\nabla\psi|^2) \\ + 2\int |d\psi|^2 + 2\int E_\lambda, \qquad E_\lambda = \langle W^+\psi, \psi\rangle + w\rho^2/3 \geqslant 0. \tag{11}$$

Nonnegativity uses the differential inequality for the parallel spinor quadratic map and $\lambda_{\min}(W^+) \geqslant -w/3$. The spinor and self-dual-form Weitzenböck identities, subtracted after testing the spinor against $\rho\Phi$, give

$$\frac{27}{8}D_0 = \int (qu^2 - f_j u^3). \tag{12}$$

For completeness, the moment argument that retains the truncation cost is as follows. Set

$$C = \int Qu^2, \quad Z = \int \sqrt{Q}\, u^3, \quad E_Q = \int Q^2.$$

Then $C - Z \geqslant 27D_0/8 \geqslant 0$, while the curvature equation gives

$$C_{\text{act}} := \int f_j^2 u^2 = 18\|F_A^+\|_2^2 \geqslant 72\pi^2 p, \quad 0 \leqslant C_{\text{act}} - C \leqslant \frac{81}{4}\int e^2 + o(1).$$

Hölder gives $C \leqslant E_Q^{1/3} Z^{2/3}$, hence

$$0 \leqslant E_Q - C \leqslant S + o(1), \qquad S := \Delta_+ + \frac{117}{4}\int e^2 \leqslant 117V\kappa.$$

If $\theta = (Z/E_Q)^{1/3}$, the nonnegative Hölder slack implies $E_Q - C \geqslant 2(C - Z)$. The associated amplitude estimate is $\int Q(u-\sqrt{Q})^2 \leqslant E_Q - C$: with probability measure $Q^2 d\mu/E_Q$, let $h = u/\sqrt{Q}$, $c = \mathbb{E}h^2$, $z_3 = \mathbb{E}h^3$, and $m = \mathbb{E}h$. Then $z_3 \leqslant c \leqslant z_3^{2/3}$, so $c \leqslant 1$; Cauchy–Schwarz gives $c^2 \leqslant mz_3 \leqslant mc$, and hence $m \geqslant c$. Thus $\mathbb{E}(h-1)^2 \leqslant 1 - c$. We have proved

$$D_0 \leqslant \frac{4S}{27} + o(1), \qquad \int Q(u - \sqrt{Q})^2 \leqslant S + o(1). \tag{13}$$

4. Produce the full calibrated form, including its Weyl direction

Because $Q \geqslant 3$, $u \leqslant 6$, and $(Q - q)^2 \leqslant e^2$, (13) yields

$$\int (q - u^2)^2 \leqslant 2(1 + 2\sqrt{3})^2 S + 2\int e^2 + o(1) \leqslant CV\kappa + o(1). \tag{14}$$

The middle inequality uses $|Q - u^2| \leqslant (1 + 2\sqrt{3})\sqrt{Q}\,|\sqrt{Q} - u|$. Together with (7), this controls $\int (u^2 - q_\beta)^2$.

Now choose the normalization, rather than assuming it:

$$c_\beta^2 = \frac{9\beta}{8(\beta+1)}, \qquad \omega = c_\beta\psi, \qquad t_\beta = 1/\beta, \qquad \alpha_\beta = \frac{9}{8(\beta+1)}.$$

It follows that

$$\|\omega\|_\infty \leqslant 3\sqrt{2}, \qquad \int (|\omega|^2 - 2)^2 \leqslant C_L V\kappa + o(1). \tag{15}$$

Define the smooth polynomial model

$$A_{\omega,t} = t\left(\frac{|\omega|^2}{3} I - \omega \otimes \omega\right).$$

The exact identity for arbitrary ψ, including its zeros, is

$$\left| W^+ - \alpha_\beta \left(\frac{\rho^2}{3} I - \psi \otimes \psi \right) \right|^2 = \frac{1}{6}(w - 2\alpha_\beta \rho^2)^2 + 2\alpha_\beta E_\lambda. \tag{16}$$

This identity uses only trace-freeness of W^+, not an eigenline selection. The tensor on its left is exactly $W^+ - A_{\omega,1/\beta}$. Also $2\alpha_\beta \rho^2 = u^2/(\beta+1)$, and

$$w - \frac{u^2}{\beta+1} = \left(w - \frac{4}{\beta} \right) - \frac{u^2 - q_\beta}{\beta+1}.$$

Equations (7), (13)–(16) give, for the same smooth solution,

$$d_\beta := \fint \left(|\operatorname{Ric} + g|^2 + |W^+ - A_{\omega,1/\beta}|^2 + (|\omega|^2 - 2)^2 \right) \leqslant C_L \kappa + o(1). \tag{17}$$

In particular the small Weyl-direction error has been proved from the actual gauge equations. It is not a hidden geometric hypothesis.

5. Supply the second-order budget

For a bounded section v of a metric bundle with compatible connection curvature $\mathscr{F}$, covariant integration by parts on a closed four-manifold gives

$$\|\nabla v\|_4^2 \leqslant 4 \|v\|_\infty \|\nabla^2 v\|_2,$$

$$\|\nabla^2 v\|_2^2 \leqslant 2\|\nabla^* \nabla v\|_2^2 + 32 \|v\|_\infty^2 \|\operatorname{Ric}\|_2^2 + 33 \|v\|_\infty^2 \|\mathscr{F}\|_2^2. \tag{18}$$

The curvature convention is $|\mathscr{F}|^2 = \sum_{ij} |\mathscr{F}_{ij}|_{\mathrm{HS}}^2$. These are the inherited Hessian identities, not a Sobolev embedding. One first interchanges Hessian slots and then commutes on ∇v; no derivative of curvature is introduced.

Equations (7), (10) give $\int f_j^2 \leqslant C_L V(1+\kappa) + o(1)$. The determinant Chern–Weil identity and (9) imply

$$\|F_A\|_2^2 = 2\|F_A^+\|_2^2 - 4\pi^2 a^2 \leqslant 4 \int f_j^2 + 4\pi^2 (-a^2)_+.$$

The curvature of the positive-spinor connection is bounded in norm by the $W^+, s, \mathring{\operatorname{Ric}}, F_A$ blocks; W^- acts trivially on this representation. Its squared integral is therefore at most $C_L V(1+\kappa+b_a)+o(1)$. The spinor equation and $|\Phi| \leqslant 2$ likewise bound $\int |\nabla_A^* \nabla_A \Phi|^2$ by $C_L V(1+\kappa) + o(1)$.

Apply (18) to Φ, and twice differentiate the parallel quadratic map $\Phi \mapsto \omega$. The resulting terms are bounded by $C(|\Phi||\nabla_A^2 \Phi| + |\nabla_A \Phi|^2)$. Finally use $\Delta_H \omega = \nabla^* \nabla \omega + (s/3)\omega - 2W^+\omega$. This proves

$$B_\omega := \fint |\Delta_H \omega|^2 \leqslant C_L(1 + \kappa + b_a) + o(1),$$

$$\boxed{(1 + B_\omega) d_\beta \leqslant C_L \kappa (1 + \kappa + b_a) + o(1).} \tag{19}$$

All the forms are smooth actual solutions. None of these estimates differentiates the smoothing weight.

6. Spectral compatibility and completion of the proof

The bounded self-dual-form compatibility theorem states that, for $\|\omega\|_\infty \leqslant K$ and bounded constant t,

$$|t|\,|1-2t| \leqslant C\sqrt{d_t(1+B_\omega+d_t)}, \tag{20}$$

where d_t is as in (17). Its proof is recorded in full in the analytic criterion, Sections 2, 3 and 6. The mechanism relevant to its use here is worth making explicit. The same Hessian estimate bounds second derivatives of $A_{\omega,t}$. The weak contracted Bianchi identity, tested against $J_t = \operatorname{div} W(A_{\omega,t})$, gives

$$⨍ |J_t|^2 \leqslant C|t|\sqrt{d_t(1+B_\omega+d_t)}.$$

If $v = |\omega|^2$, the polynomial pointwise contraction gives

$$t^2 v(|\nabla\omega|^2 + |d\omega|^2) \leqslant |J_t|^2 + \tfrac{4}{3}t^2|dv|^2, \qquad ⨍ |dv|^2 \leqslant C\sqrt{d_t(1+B_\omega+d_t)}.$$

Testing the self-dual Hodge equation against $v\omega$ gives an algebraic term $16t^2(1-2t)/3$; all its derivative terms are controlled by the preceding two displays. The zeroth-order errors cost at most $Ct^2\sqrt{d_t}$. Thus

$$\left|\frac{16}{3}t^2(1-2t)\right| \leqslant 7⨍ |J_t|^2 + Ct^2\sqrt{d_t(1+B_\omega+d_t)}.$$

Division by $|t|$, with the case $t = 0$ separate, proves (20). The argument neither differentiates the Ricci error nor assumes $d\omega = 0$.

Substitute the produced form, with $t = 1/\beta$, into (20). Using (17)–(19), then letting its smoothing allowance tend to zero,

$$\frac{(\beta-2)^2}{\beta^4} \leqslant C_L\kappa(1+\kappa+b_a). \tag{21}$$

Absorb $\beta^4 \leqslant L^4$ into the constant. When $\kappa \leqslant 1$, the right side is bounded by $2C_L(1+b_a)\kappa$. When $\kappa > 1$, (2) is immediate after choosing $c_L \leqslant (L-2)^{-2}$. This proves (2).

For (3), Chern–Weil gives

$$4\pi^2 T = \int \left(2|W^+|^2 + s^2/24 - |\operatorname{Ric}^\circ|^2/2\right) \geqslant -\tfrac{1}{2}Vr.$$

Hence $a^2 \geqslant T$ implies $b_a \leqslant r/(8\pi^2)$. For $\kappa \leqslant 1$ this is bounded by $1/(8\pi^2)$; for $\kappa > 1$ use the same trivial split. Equation (4) is (3) with $r = 0$, after multiplying by $\beta V/(\beta+1)$.

7. The allowed endpoint without a scalar-curvature band

The same construction has a structural consequence even when the parameter approaches the allowed endpoint. It is not obtained by declaring that endpoint forbidden.

Let H be the harmonic self-dual representative of a_g^+, with convention $\operatorname{proj}_{\mathscr{H}^+}(iF_A^+) = 2\pi H$. Set

$$\zeta_\beta = -\frac{3\pi\beta}{\beta+1}H.$$

Theorem 2 (actual harmonic alignment). Under the hypotheses of Theorem 1, if $\kappa \leqslant 1$, the same smooth forms can be chosen with arbitrarily small allowance $\eta > 0$ so that

$$⨍ |\omega-\zeta_\beta|^2 \leqslant C_L\kappa + \eta, \qquad ⨍ |\nabla\omega|^2 \leqslant C_L\sqrt{\kappa(1+\kappa+b_a)} + \eta. \tag{22}$$

The amplitude, polynomial Weyl fit and Hodge-Laplacian bounds remain (15), (17) and (19). In particular, at $\beta = 2$, the harmonic target is $-2\pi H$, and $\kappa = r + 144\pi^2 D_{\mathrm{LB}}/V$. This conclusion needs neither a scalar band nor a scalar spectral gap.

Proof. Retain the actual curvature moment from Section 3. Since $C \leqslant E_Q$, its Hodge remainder satisfies

$$18\|iF_A^+ - 2\pi H\|_2^2 = C_{\mathrm{act}} - 72\pi^2 p \leqslant S + o(1).$$

The curvature equation therefore gives

$$\|f_j\psi + 4\sqrt{2}\pi H\|_2^2 \leqslant 4S/9 + o(1). \tag{23}$$

Furthermore

$$\int (f_j - \sqrt{q_\beta})^2 \leqslant 2\int (f_j - \sqrt{Q})^2 + 2\int (\sqrt{Q} - \sqrt{q_\beta})^2 \leqslant C_L V\kappa + o(1).$$

Here $(f_0 - \sqrt{Q})^2 \leqslant f_0^2 - Q \leqslant 9e^2/16$, and the square-root map is uniformly Lipschitz on $[3,\infty)$. Use $|\psi| \leqslant 4$ and multiply (23) by c_β^2/q_β. The identity $4\sqrt{2}\pi c_\beta/\sqrt{q_\beta} = 3\pi\beta/(\beta+1)$ proves the first estimate in (22).

For the second, the derivative conclusion of the bounded-form theorem on $1/L \leqslant t \leqslant 1/2$ is $⨍ |\nabla\omega|^2 \leqslant C_L\sqrt{d_t(1 + B_\omega + d_t)}$ when $d_t \leqslant 1$. If κ is small, (17) ensures this. For the remaining compact range of $\kappa \leqslant 1$, the integrated Hodge equation and the bound on B_ω give $⨍ |\nabla\omega|^2 \leqslant C_L\sqrt{1 + b_a}$; enlarging the constant recovers (22). All allowances are taken small simultaneously for the same solution.

The harmonic form need not have a uniform maximum bound. Thus (22) does not license replacing ω by ζ_β inside the quadratic Weyl model. It does give a precise global spectral conclusion: if $\Pi_{>\lambda}$ is the spectral projection of the rough Laplacian on self-dual forms, then for every $\lambda > 0$,

$$⨍ |\Pi_{>\lambda}\zeta_\beta|^2 \leqslant C_L\left[\kappa + \frac{\sqrt{\kappa(1 + \kappa + b_a)}}{\lambda}\right]. \tag{24}$$

Indeed, project $\zeta_\beta - \omega$ and use $\|\Pi_{>\lambda}\omega\|_2^2 \leqslant \lambda^{-1}\|\nabla\omega\|_2^2$, then let the allowance tend to zero. This controls an actual specified harmonic class; it does not assert global Kähler topology or exact metric correction.

At $\beta = 2$, there is a stronger direct estimate, independent of b_a. In the notation of (11), the retained Hodge identity is

$$\int |\nabla\psi|^2 = 2\int |d\psi|^2 + 2\int E_\lambda - \tfrac{1}{3}\int (s + 2w)\rho^2.$$

Since $\rho \leqslant 4$, $\int (s + 2w)^2 \leqslant 2\mathfrak{D}_2$, and $c_2^2 = 3/4$, this gives simultaneously

$$⨍ |\nabla\omega|^2 \leqslant G(\kappa) + \eta, \quad G(\kappa) = 13\kappa + 4\sqrt{2\kappa}, \quad ⨍ |d\omega|^2 \leqslant 13\kappa/2 + \eta. \tag{25}$$

The preceding harmonic-alignment estimate can likewise be made explicit:

$$⨍ |\omega + 2\pi H|^2 \leqslant 49\kappa + \eta. \tag{26}$$

For (26), (6) gives $⨍(q - 6)^2 \leqslant 18\kappa$. The cutoff estimate implies $⨍(f_j - \sqrt{6})^2 \leqslant 9\kappa + o(1)$, and (23) then costs at most $4 \cdot 9\kappa + S/(9V) \leqslant 49\kappa$. Thus in (24) the endpoint right side can be replaced by $98\kappa + 2G(\kappa)/\lambda$, with no square-cost hypothesis. The endpoint application gives this statement and its spectral consequences separately.

8. What is strengthened, and what is not

At $\beta = 2$,

$$\mathfrak{D}_2 = 144\pi^2 D_{\mathrm{LB}}, \qquad D_{\mathrm{LB}} = \frac{1}{4\pi^2}\int (s^2/24 + |W^+|^2/2) - p/2.$$

The classical equality set is negative cscK, compatible with the selected Spin^c structure, not necessarily Einstein. Equation (2) does not turn its zero deficit into an unconditional Ricci estimate.

At $\beta = 8$, Chern–Weil gives

$$\mathfrak{D}_8 = 108\pi^2(T - 2p/3) + \frac{27}{2}\int |\operatorname{Ric}^\circ|^2. \tag{27}$$

Thus the old topological calibration is one member of the family. For other parameters, (2) controls a different scalar–Weyl functional, with its own genuine angular calibration supplied in Section 4. This is a functional extension of the method, not a second discovery of the old 1/3 blow-up obstruction.

An exact negative Kähler–Einstein metric with canonical class has $s = -4, w = 2, p = V/(2\pi^2)$. Direct substitution yields

$$\frac{\mathfrak{D}_\beta}{V} = \frac{4(\beta - 2)^2}{\beta}. \tag{28}$$

The quadratic vanishing at the allowed endpoint is consequently the correct order; the theorem does not claim an optimal coefficient.

For $0 < \beta \leqslant 2$, the classical scalar and scalar–Weyl bounds already provide a stronger elementary bound. If $x = \|s\|_2, y = \|w\|_2$, then $x \geqslant \sqrt{32\pi^2 p}$ and $x + y \geqslant \sqrt{72\pi^2 p}$. Minimizing $x^2 + \beta y^2$ on these constraints gives $\int (s^2 + \beta w^2) \geqslant 8\pi^2 p(4 + \beta)$. Thus this range does not need the new compatibility argument. For $\beta > 2$, that scalar constraint is inactive at the classical relaxed minimizer. The additional obstruction in (2) is differential, through weak Bianchi and the actual bounded form.

For the Yau26 negative input, $r \leqslant 4\delta_-(g)^2$. Under $a^2 \geqslant T$, (3) therefore gives the usable estimate

$$4\delta_-(g)^2 + \mathfrak{D}_\beta/V \geqslant c'_L(\beta - 2)^2. \tag{29}$$

To obtain a new purely topological application one must additionally force $\mathfrak{D}_\beta/V$ small from independently justified topological or geometric data. No such new topology is asserted here. Neither (29) nor the allowed-root form supply proves existence of an Einstein metric on an arbitrary smooth manifold.

Sources and verification

- C. LeBrun, *Ricci Curvature, Minimal Volumes, and Seiberg–Witten Theory*, Theorem 2.3 and Propositions 3.1, 4.1: the classical curvature inequalities and cscK equality set.
- C. LeBrun, *Four-manifolds, curvature bounds, and convex geometry*, Lemma 3.7: weighted equations.
- Integral truncated-weight proof, Sections 2–4: the inherited cutoff and moment estimates, reproduced above.
- Bounded-form spectral theorem: the analytic compatibility theorem, including its polynomial contractions.
- Scalar-band supply: the prior allowed cscK branch, which is not itself a forbidden-spectrum application.

Exact Einstein saturation forcing Kähler–Einstein structure already appears in LeBrun's Corollary 2.5; it is not a new classification here. The quantitative persistence under integral Ricci error, and the actual all-scalar form-production estimates, are the assertions at issue.

The parameterized all-scalar functional estimate (2) is the claim proved in this note. Its finite algebra is checked separately; those checks do not replace the gauge-solvability hypothesis or the analytic proofs. No comprehensive priority claim over later literature is made.